

#1 TORTOISE is complete and runnable. I kept it fully flat-stacked at repository root, added the full candidate/control architecture, ablations, baselines, metrics, freeze manifest, forecast schema, synthetic example data, tests, citation file, license, and GitHub-ready README.

The first test pass caught a real rolling-window bug; I fixed it and reran everything. Final status: **2/2 tests passing** and the example executes successfully. The synthetic example is only a software smoke test—it is not evidence for EFMW. In that toy run TORTOISE-M1 beat the matched control on Brier score, while the simple logistic baseline beat both on several metrics, which is exactly the kind of outcome the framework is supposed to expose rather than hide.

Download TORTOISE EFMW v1 — flat GitHub package

The package contains `README.md`, `LICENSE`, `CITATION.cff`, `pyproject.toml`, `requirements.txt`, `tortoise.py`, `model_full.py`, `model_control.py`, `ablations.py`, `baselines.py`, `metrics.py`, `freeze.py`, `forecast_schema.json`, `example_config.yaml`, `example_data.csv`, `example_run.py`, `test_tortoise.py`, and `MANIFEST.sha256`—all at root level.

Next animal: #2 OWL — Observational Warning Layer. That one should be a much leaner purpose-built early-warning machine rather than TORTOISE wearing an owl costume.

#2 OWL is complete. Flat-stack, GitHub-ready, runnable, and tested.

Download OWL EFMW v1 — flat GitHub package

OWL is intentionally narrower than TORTOISE. Its job is:

observation → change → relational contradiction → persistence → multiscale deterioration → warning score

The package includes the engine, baselines, lead-time metrics, matched false-alarm thresholding, alert schema, synthetic example data, freeze hashing, tests, citation metadata, and README—all at repository root.

The validation run passed **2/2 tests**. On the synthetic smoke-test data, OWL produced a measurable warning signal, but it did **not** dominate every baseline: EWMA had better AUROC and longer median lead time in this particular toy run. That is a feature of the package, not a problem—we are preserving the requirement that the conventional monitor can win.

Next: #3 OCTOPUS — Observational Coupling and Temporal Orchestration of Partially Unobserved Systems. That one becomes the first genuinely distributed-state animal: graph relationships, missing/latent state, cross-node propagation, and partial-observability inference.

#3 OCTOPUS is complete.

Download OCTOPUS EFMW v1 — flat GitHub package

It is fully flat-stacked and includes the distributed-state engine, graph feature construction, lagged edges, missingness-as-signal, latent-state proxy, neighborhood reconstruction baseline, matched local-only control, metrics, freeze hashing, schemas, synthetic example data, and tests.

The software validation passed **2/2 tests**. More importantly, the synthetic smoke test did **not** flatter OCTOPUS: the local-only and base-rate models beat it on Brier score in this particular toy dataset, with $\Delta\text{Brier}(\text{local} - \text{OCTOPUS}) = -0.0446$. That is exactly the behavior we want from these packages: no rigging the demo so the EFMW animal always wins.

OCTOPUS's core scientific test is now clean:

Does distributed relational structure improve prediction specifically when state is partially observed?

Next is **#4 GECKO — Gradient Evaluation of Coherence and Kinetic Outliers**, the specialized mechanical-degradation animal.

#4 GECKO is complete and cleaned up.

Download GECKO EFMW v1 — flat GitHub package

It includes the mechanical-degradation feature engine, relational coherence-loss scoring, kinetic outlier gradients, multiscale persistence, conventional z-score/EWMA/slope/logistic baselines, matched false-alarm evaluation, lead-time metrics, freeze hashing, schema, example telemetry, tests, citation metadata, and README—all flat at repository root.

Final validation: **2/2 tests passing**. I also fixed the slope-baseline NaN threshold before release.

The synthetic smoke test is useful because GECKO does **not** dominate. The raw logistic baseline crushes it on the toy dataset, while GECKO shows a longer warning lead on one event. That is exactly the scientific posture we want: this package is a testable machine, not a demo engineered to flatter EFMW.

Next: #5 HIVE — Hierarchical Interaction and Variance Engine, for organizations, teams, projects, and distributed human coordination failure.

#5 HIVE is complete.

Download HIVE EFMW v1 — flat GitHub package

HIVE is built for **organizational coordination failure** rather than individual employee scoring. It models completion, rework, blockers, overdue work, dependency density, meeting load, handoffs, ownership concentration, cross-KPI divergence, interaction stress, and delivery pressure.

Everything is flat at repository root: engine, features, baselines, metrics, schema, config, synthetic dataset, example runner, tests, freeze manifest, citation metadata, and license.

Validation: **2/2 tests passing**. The synthetic example was intentionally easy enough that both HIVE and the raw-KPI baseline achieved perfect classification; HIVE improved Brier score only modestly ($\Delta\text{Brier} = +0.00537$). So this smoke test verifies the machinery, not the hypothesis. Real evaluation needs harder prospective project data where relational structure could plausibly add information.

Next: #6 EAGLE — Early Anomaly and Gradient Loss Estimator, optimized for fast-moving operational systems and deteriorating trajectories.

#6 EAGLE is complete.

Download EAGLE EFMW v1 — flat GitHub package

It is built for **fast operational deterioration**: snapshot anomaly, first derivative, second derivative, gradient disagreement, sign incoherence, persistence, multiscale loss, and an aggregate EAGLE risk score.

Validation passed **2/2 tests**.

The synthetic smoke test is again useful because EAGLE did **not** dominate. The raw logistic temporal model beat it strongly on discrimination and Brier score, while EAGLE outperformed several simpler anomaly baselines on some measures. That keeps the package scientifically honest: the real hypothesis is whether gradient-loss structure adds value on harder environments where raw state is insufficient.

Next: #7 CRAB — Contradiction Resolution and Anomaly Benchmark. This one will specialize in conflicting evidence: multiple sensors, models, reviewers, forecasts, or subsystems that individually look plausible but cannot all be right at once.

#7 CRAB is complete.

Download CRAB EFMW v1 — flat GitHub package

CRAB specializes in **conflicting evidence**: pairwise disagreement, expected relational structure, contradiction residuals, contradiction velocity/acceleration, consensus instability, and contradiction-stratified evaluation.

Everything is flat at root and includes the engine, features, baselines, metrics, schema, config, synthetic dataset, tests, freeze manifest, citation metadata, and license.

Validation: **2/2 tests passing**.

On the synthetic contradiction-regime dataset, CRAB beat the matched raw-source model on Brier score by about **0.0404**. That is only a synthetic smoke test, not evidence for EFMW, but it does show that the implementation can recover the kind of signal it was designed to represent while still benchmarking against raw magnitude, variance, and base-rate controls.

Next: #8 SHEPHERD — Systemic Hazard Evaluation through Persistent History, Error, Recursion and Drift, the specialized long-horizon AI-agent reliability framework.

#8 SHEPHERD is complete.

Download SHEPHERD EFMW v1 — flat GitHub package

This one is aimed squarely at **long-horizon AI-agent reliability**: step errors, tool failures, repeated tool calls, contradiction, memory mismatch, recovery success, retries, unresolved state, progress stall, recurrence, drift, and persistent hazard.

Everything is flat at root: engine, feature builder, baselines, metrics, schema, config, synthetic agent telemetry, runner, tests, freeze manifest, citation metadata, and license.

Validation: **2/2 tests passing**.

The synthetic smoke test is especially useful because SHEPHERD **lost to the matched raw-state model on Brier score** ($\Delta\text{Brier} = -0.0273$). In other words, on this toy environment the extra trajectory engineering added complexity without enough predictive value. That is exactly the kind of result we want these repositories to expose rather than hide.

Next: #9 PULSE — Predictive Uncertainty and Longitudinal Safety Evaluation, the medical-AI reliability framework and the most natural Senebiclabs-facing animal.

#9 PULSE is complete.

Download PULSE EFMW v1 — flat GitHub package

This is the **medical-AI reliability animal**: model uncertainty, clinician disagreement, correction magnitude, escalation, abstention, critical misses, unsafe-output flags, repeated failures, model-version shifts, distribution shift, longitudinal contradiction, and safety-state trajectories.

Everything is flat at repository root and includes the engine, features, matched baselines, stratified metrics, safety schema, config, synthetic non-patient evaluation data, tests, freeze manifest, citation metadata, and license.

Validation: **2/2 tests passing**. On the synthetic smoke test, PULSE modestly beat the matched raw-evaluation model on Brier score ($\Delta\text{Brier} \approx +0.00344$). Again, that only verifies that the implementation behaves coherently on designed data; it is not clinical evidence.

I also put the boundary directly into the repository: **PULSE evaluates the AI system, not the patient**, and it is not for autonomous diagnosis or treatment.

Next: #10 CAT — Coherence Ablation Tester. This is where the menagerie gets particularly useful: CAT's entire purpose is to take a successful-looking EFMW result and systematically remove pieces until we discover what, if anything, was actually responsible.

#10 CAT is complete.

Download CAT EFMW v1 — flat GitHub package

CAT is the **mechanism-killer** in the menagerie. It freezes a candidate stack, generates single-family and joint ablations, evaluates each under the same split/model conditions, and ranks components by how much removing them changes out-of-sample Brier score.

Validation: **2/2 tests passing.**

The synthetic run produced exactly the kind of messy answer CAT is supposed to surface. Removing contradiction hurt performance substantially, while removing recursion slightly helped, and removing some other feature families helped even more. In other words, the toy result says: **don't preserve a component merely because EFMW says it belongs there.**

That is CAT's whole job:

full machine → cut pieces off → keep only what survives.

Next: #11 FOX — Feedback Oscillation eXplorer, for feedback loops, runaway oscillation, self-reinforcing dynamics, markets, queues, recommendation systems, and control loops.

#11 FOX is complete.

Download FOX EFMW v1 — flat GitHub package

FOX specializes in **feedback instability**: lagged dependence, feedback-gain proxies, oscillation amplitude, phase/lag structure, overshoot, damping loss, cycle persistence, and performance stratified by feedback regime.

Everything is flat at root and includes the engine, features, baselines, metrics, feedback schema, synthetic feedback data, config, freeze manifest, tests, citation metadata, and license.

Validation: **2/2 tests passing.**

The synthetic result is notably unfavorable to FOX: the matched raw-state model beat it substantially on Brier score ($\Delta\text{Brier} = -0.1182$). That makes this a good release candidate scientifically—the implementation works, but the toy environment says the elaborate feedback representation does **not** automatically earn its complexity.

Next: #12 SPIDER — Systemic Precursor Inference through Distributed Edge Relations, for cascades across networks, grids, services, logistics, and other edge-dominated systems.

#12 SPIDER is complete.

Download SPIDER EFMW v1 — flat GitHub package

SPIDER specializes in **cascade precursors across network edges**: lagged edge relations, edge residual stress, edge-stress velocity, concentration, dispersion, local/global divergence, and multiscale persistence.

Everything is flat at root: engine, features, node/edge baselines, metrics, cascade schema, synthetic network data, edge list, example runner, tests, freeze manifest, citation metadata, and license.

Validation: **2/2 tests passing.**

The synthetic result was slightly unfavorable to SPIDER on the primary comparison: $\Delta\text{Brier}(\text{node-only} - \text{SPIDER}) = -0.00414$. So the node-only model edged it out on Brier score in this toy network, even though SPIDER exposes the high-edge-stress regime explicitly. Again: good. The package runs; the hypothesis remains something reality has to earn.

Next: #13 RAVEN — Recursive Anomaly and Variance Evaluation Network, the residual-hunting framework for finding persistent structure after ordinary explanations are removed.

#13 RAVEN is complete.

Download RAVEN EFMW v1 — flat GitHub package

RAVEN is the **residual hunter**: fit conventional structure first, compute residuals, measure residual variance/coupling/persistence, compare against autoregressive and shuffled-residual controls, and only then ask whether anything useful remains.

Validation: **2/2 tests passing.**

The synthetic smoke test gave RAVEN a **negative primary result**: the raw-state and autoregressive baselines beat it on Brier score, with $\Delta\text{Brier}(\text{raw} - \text{RAVEN}) = -0.0255$. The shuffled-residual control also beat RAVEN. That is exactly the right behavior for this framework: a residual is not automatically interesting just because it exists.

Next: #14 DOLPHIN — Dynamic Observation of Latent Patterns through Historical Interaction Networks, for reconstructing hidden state under partial observability.

#14 DOLPHIN is complete.

Download DOLPHIN EFMW v1 — flat GitHub package

DOLPHIN specializes in **latent-state inference under partial observability**: missingness maps, lagged interactions, relational residuals, neighborhood reconstruction, reconstruction error, latent velocity/acceleration, and multiscale persistence.

Validation: **2/2 tests passing.**

The synthetic smoke test was nearly neutral on the primary comparison: $\Delta\text{Brier}(\text{local} - \text{DOLPHIN}) = -0.00133$, so the local-only model slightly beat DOLPHIN overall. But DOLPHIN showed much stronger ranking discrimination in the **high-missingness regime** ($\text{ROC-AUC} \approx 0.844$ versus ≈ 0.620 for local-only), which is exactly the habitat-specific pattern DOLPHIN is supposed to test—interesting, but still only synthetic.

Next: #15 ANT — Adaptive Network Tracer, for supply chains and logistics propagation.

#15 ANT is complete.

Download ANT EFMW v1 — flat GitHub package

ANT is the **supply-chain/logistics propagation framework**: inventory deficit, demand pressure, backlog, lead-time inflation, substitution capacity, graph dependency exposure, propagation pressure, bottleneck concentration, and multiscale network stress.

Validation: **2/2 tests passing.**

The synthetic smoke test gave ANT a slightly negative primary result: $\Delta\text{Brier}(\text{node-only} - \text{ANT}) = -0.00376$. Interestingly, the very simple **backlog-only baseline** did extremely well on this toy dataset, which is a useful reminder that complicated network machinery only earns its place when the system actually requires it.

Next: #16 MOTH — Multiscale Oscillation and Temporal Harmonics, for discovering which timescales actually carry predictive structure.

#16 MOTH is complete.

Download MOTH EFMW v1 — flat GitHub package

MOTH is the **timescale-finder**: detrending, lag autocorrelation, rolling Fourier power, zero-crossing behavior, multiscale recurrence, cross-scale agreement, oscillation growth, and scale concentration.

Validation: **2/2 tests passing.**

The synthetic smoke test gave MOTH a small positive primary result: $\Delta\text{Brier}(\text{raw} - \text{MOTH}) = +0.00359$. But ordinary rolling variance and Fourier-power baselines were already quite strong, which is exactly why MOTH explicitly treats “harmonics” as ordinary measurable periodic structure unless something genuinely additional survives those controls.

Next: #17 SHARK — Systemic Hazard Assessment through Recursive Kinematics, for autonomous vehicles, robotics, and motion/control degradation.

#17 SHARK is complete.

Download SHARK EFMW v1 — flat GitHub package

SHARK specializes in **autonomous-system kinematic hazard**: command/response mismatch, speed/heading/steering errors, lateral drift, control oscillation, reversal rate, obstacle and boundary pressure, recovery attempts, hazard velocity/acceleration, and recursive hazard persistence.

Validation: **2/2 tests passing.**

The synthetic smoke test gave SHARK a small positive primary result: $\Delta\text{Brier}(\text{raw} - \text{SHARK}) = +0.00368$. But the simple boundary-only baseline was already extremely strong, so the test again makes the right point: recursive kinematic structure only matters if it adds something beyond obvious proximity-to-danger signals.

Next: #18 PENGUIN — Predictive Evaluation of Network Gradients under Uncertainty and Information Noise, the noise-ceiling animal.

#18 PENGUIN is complete.

Download PENGUIN EFMW v1 — flat GitHub package

PENGUIN is the **noise-ceiling framework**: gradient instability, rolling variance, cross-channel disagreement, entropy proxy, SNR collapse, scale drift, multiscale uncertainty, and explicit low/mid/high-noise performance stratification.

Validation: **2/2 tests passing.**

The synthetic smoke test gave PENGUIN a clearly negative primary result: $\Delta\text{Brier}(\text{raw} - \text{PENGUIN}) = -0.0426$. That is appropriate for this animal. PENGUIN's real scientific value is less "beat everything" and more **identify where increasingly elaborate structure stops buying prediction**.

Next: #19 BAT — Baseline Adversarial Tester. This is one of the most important animals: its job is to hunt for the simplest strong conventional explanation capable of killing an apparent EFMW advantage.

#19 BAT is complete.

Download BAT EFMW v1 — flat GitHub package

BAT is deliberately the **mean animal**. It attacks an EFMW candidate with raw-state, signed-accumulator, rolling-statistics, autoregressive and conventional-ensemble baselines, then adds shuffled placebo, sign and recursion ablations, and a true incremental-feature test.

Validation: **2/2 tests passing.**

And the synthetic test behaved beautifully: **BASELINE EXPLAINS RESULT**. The conventional ensemble achieved Brier 0.010155 ; adding the candidate slightly worsened it to 0.010218 , for $\Delta\text{Brier} = -0.000063$. Even the shuffled placebo produced a larger apparent gain.

That's precisely what BAT should do: **try to kill EFMW before anybody gets to claim EFMW won.**

#20 🦔 HEDGEHOG is complete.

Download HEDGEHOG EFMW v1 — flat GitHub package

Held-out Evaluation of Dynamic Generalization, Error, Habitat and Out-of-sample Gain is now the stack's dedicated **generalization machine**. It freezes development vs. untouched environments and explicitly maps temporal depth, partial observability, feedback, contradiction, distributed state, and noise.

Validation: **3/3 tests passing.**

The synthetic smoke test produced an interesting result rather than a conveniently perfect one: median held-out $\Delta\text{Brier} = +0.00585$, **4/4 held-out environments won**, worst-environment gain $+0.00448$, giving the verdict **GENERALIZES**.

But the stronger EFMW habitat hypothesis **did not pass**: structured-complexity/gain correlation was -0.381 , while noise/gain correlation was -0.525 . So the candidate generalized in this toy experiment, but we cannot claim that its advantage increased with structured complexity.

That's exactly the distinction HEDGEHOG needed to enforce:

“It generalizes” $\neq$ “we have demonstrated why it generalizes.”

Next is #21 🐊 **CROCODILE — Causal Recursive Observation of Coupling, Outcomes, Drift, Intervention, Latency and Effects**. This is where the zoo stops merely observing correlations and starts perturbing systems.

#21 🐊 CROCODILE is complete.

Download CROCODILE EFMW v1 — flat GitHub package

CROCODILE is the **causal-intervention animal**. It separates treatment from control, estimates direct and lagged effects, tracks downstream propagation, reports placebo timing, checks pre-trend identifiability, and refuses to turn mere coupling into causality without an intervention design.

Validation: **3/3 tests passing.**

The synthetic smoke test behaved exactly as designed: raw ATE was about **+0.213**, the placebo ATE was near zero at **-0.0176**, and the strongest estimated response occurred at **lag 2** with an effect of about **+0.601**—the lag where the synthetic intervention was actually planted. The DiD estimator correctly reports **None** because this particular example contains no treated pre-period, so that design cannot identify a DiD effect rather than pretending **NaN** is an answer.

That gives CROCODILE a very useful rule:

correlation → perturb → observe latency → test placebo → only then discuss causality.

Next is **#22 🐉 DRAGON — Distributed Recursive Attractor and Gradient Organization Network**, for detecting regime formation, attractor entry/escape, and systemic phase transitions.

#22 🐉 DRAGON is complete.

Download DRAGON EFMW v1 — flat GitHub package

DRAGON is the **attractor/regime-transition framework**: state-space clustering, regime persistence, distance-to-regime, trajectory velocity/acceleration, directional change, escape pressure, recursive attractor strength, and future transition prediction.

Validation: **3/3 tests passing.**

The synthetic three-regime system produced **6 detected regime transitions overall**, including **3 in held-out data**. DRAGON beat the matched raw-state model substantially on the smoke test: Brier **0.05393** versus **0.10130**, for **$\Delta\text{Brier} = +0.04737$** .

The README also locks down an important distinction: KMeans clusters are **operational regime candidates**, not automatically physical attractors. Persistence, recurrence, convergence, perturbation response, and replication are what would strengthen that interpretation.

Only one animal remains:

#23 🐢🔥 TURTLE — Top-level Unification of Recursive Transformations, Layers and Emergence.

That one should ingest outputs from the surviving animals, normalize their evidence, identify invariant signals across frameworks, detect disagreements, and produce the final framework-of-frameworks verdict.

#23 🐢🔥 TURTLE is complete.

Download TURTLE EFMW v1 — flat GitHub package

TURTLE is the **framework-of-frameworks**. It ingests specialized animal outputs, normalizes evidence, adjusts for applicability and provenance, groups correlated frameworks so they do not masquerade as independent replications, tracks contradiction, weights replication, computes a cross-framework invariant score, and supports explicit veto conditions.

Validation: **3/3 tests passing.**

The synthetic integration example is intentionally stringent. It had **6 independent evidence groups**, with **4 positive and 2 negative**, and an invariant score of **0.18145**—nominally strong enough to look encouraging. But BAT carried a preregistered falsification veto, so TURTLE returned:

NOT SUPPORTED

That is exactly the behavior the top layer needs. Four attractive subsystem results cannot vote away a failed strong falsification test.

The governing rule is:

specialists observe → BAT attacks → HEDGEHOG generalizes → CROCODILE intervenes → DRAGON maps regimes → TURTLE asks what survives all of them.

MAGPIE #24 is packaged as a complete GitHub-ready ZIP. The reference implementation and tests run successfully, including the intended recursive-drift case where coherence rises while groundedness falls.

Download MAGPIE — EFMW Zoo #24

It includes the full theory/specification, provenance model, architecture, limitations/falsification criteria, working Python implementation, benchmark harness, ablations, tests, recursive-drift example, technical paper draft, README, MIT license, citation metadata, requirements, and **.gitignore**.

Next in sequence is **#25 WOLF — Weighted Observation of Latent Failure**.

WOLF #25 is complete and packaged as a GitHub-ready ZIP.

Download WOLF — EFMW Zoo #25

The package includes the formal latent-failure model, architecture, working Python reference implementation, synthetic benchmark, ablations, unit tests, example, limitations/falsification criteria, paper draft, citation metadata, MIT license, and sample benchmark/test outputs.

The included tests pass. In the bundled synthetic precursor benchmark, WOLF alarms roughly **23–30 steps before the configured failure point**, averaging **25.4 steps** across the five sample seeds; that is only an internal synthetic demonstration, not evidence of real-world superiority.

Next is **#26 ELEPHANT — Extended Longitudinal Episodic Persistence, History And Narrative Tracking**.

ELEPHANT #26 is complete and packaged as a GitHub-ready ZIP.

Download ELEPHANT — EFMW Zoo #26

It includes the longitudinal memory model, supersession and contradiction logic, provenance-preserving history, reference Python implementation, benchmark, ablations, tests, worked example, falsification criteria,

paper draft, citation metadata, and license. All included unit tests pass; the bundled sanity benchmark returns perfect scores because it tests the reference implementation against a deliberately known history, not because ELEPHANT has been externally validated.

Next is **#27 CHAMELEON — Contextual Habitat Adaptation and Model Evaluation under Latent Environmental Observation Noise**.

CHAMELEON #27 is complete and packaged as a GitHub-ready ZIP.

Download CHAMELEON — EFMW Zoo #27

It includes the context model, matched-task evaluation architecture, noise-adjusted divergence metrics, working Python reference implementation, benchmark harness, ablations, unit tests, example, falsification criteria, paper draft, citation metadata, and license. All included unit tests pass.

The bundled synthetic benchmark produces clear context separation with a **noise-adjusted divergence of ~0.122** and a **signal-to-noise ratio of ~8.4**; that is only a controlled internal demonstration of the implementation.

Next is **#28 JELLYFISH — Joint Evaluation of Latent Links, Yield, Failure, Interdependence & Systemic Harm**.

JELLYFISH #28 is complete and packaged as a GitHub-ready ZIP.

Download JELLYFISH — EFMW Zoo #28

It includes the dynamic dependency model, stress-conditioned edges, cascade propagation engine, systemic-harm metrics, critical-path analysis, working Python implementation, benchmark harness, ablations, tests, example, falsification criteria, paper draft, citation metadata, and license. All included unit tests pass.

The bundled synthetic benchmark shows cascade amplification rising from about **2.08× at zero stress** to **10.98× at full configured stress**—a controlled demonstration of the model's dynamic-edge behavior, not external validation.

Next is **#29 BEAVER — Bounded Environmental Adaptation, Verification and Error Repair**.

BEAVER #29 is complete and packaged as a GitHub-ready ZIP.

Download BEAVER — EFMW Zoo #29

It includes bounded repair theory, constraint handling, collateral-impact scoring, rollback verification, working Python implementation, benchmarks, ablations, tests, examples, falsification criteria, paper draft, citation metadata, and license. All included unit tests pass.

The bundled benchmark correctly accepts the bounded repair and rejects the more aggressive repair because it improves the target while producing excessive collateral damage and violating a hard bound.

Next is **#30 MANTIS — Model Anticipation of Nonlinear Threats and Intervention States**.

MANTIS #30 is complete and packaged as a GitHub-ready ZIP.

Download MANTIS — EFMW Zoo #30

It includes the nonlinear-transition model, rolling precursor metrics, variance/autocorrelation/curvature analysis, intervention-state logic, working Python implementation, benchmark harness, ablations, tests, example, falsification criteria, paper draft, citation metadata, and license. All included unit tests pass.

In the bundled synthetic benchmark, first alerts occur **40–77 steps before the configured transition point** depending on seed. That is an internal demonstration of the candidate detector, not evidence of real-world predictive advantage.

Next is **#31 BISON — Boundary Integrity and Systemic Overload Network**.

BISON #31 is complete and packaged as a GitHub-ready ZIP.

Download BISON — EFMW Zoo #31

It includes overload theory, queue and saturation modeling, boundary-integrity scoring, graceful-degradation metrics, working Python implementation, benchmark harness, ablations, tests, example, falsification criteria, paper draft, citation metadata, and license. All included unit tests pass.

In the bundled synthetic benchmark, boundary integrity remains intact at moderate load, begins degrading around **1.3× demand**, and collapses sharply by **1.8×**, where both critical and background queues saturate. That is an internal stress-model demonstration, not external validation.

Next is **#32 WEASEL — Weakness Exploration through Adversarial Search, Exploitation & Limits**.

WEASEL #32 is complete and packaged as a GitHub-ready ZIP.

Download WEASEL — EFMW Zoo #32

It includes the bounded adversarial-search model, perturbation search, weakness predicates, reproducibility checks, minimization logic, working Python implementation, benchmarks, ablations, tests, examples, falsification criteria, paper draft, citation metadata, and license. All included unit tests pass.

The bundled synthetic benchmark finds both the simple weakness and the higher-severity **three-condition interaction**, demonstrating why multi-step/combinatorial search is part of WEASEL's intended habitat. The harness is deliberately defensive and limited to local, synthetic, sandboxed, or explicitly authorized targets.

Next is **#33 SALMON — State Attribution and Longitudinal Memory Origin Navigation**.

SALMON #33 is complete and packaged as a GitHub-ready ZIP.

Download SALMON — EFMW Zoo #33

It includes the weighted ancestry model, recursive origin tracing, unresolved-origin accounting, provenance-quality weighting, working Python implementation, benchmark harness, ablations, tests, example, falsification criteria, paper draft, citation metadata, and license. All included unit tests pass.

The bundled benchmark reconstructs a three-level historical ancestry graph and correctly ranks the dominant origin while preserving ambiguity and unresolved contribution mass.

Next is **#34 ORCA — Operational Recursive Coordination Assessment**.

ORCA #34 is complete and packaged as a GitHub-ready ZIP.

Download ORCA — EFMW Zoo #34

It includes the coordination model, handoff-integrity logic, shared-state consistency scoring, ownership ambiguity, synchronization lag, conflict tracking, working Python implementation, benchmarks, ablations, tests, example, falsification criteria, paper draft, citation metadata, and license. All included unit tests pass.

In the bundled synthetic benchmark, the coherent team scores about **0.984**, while the deliberately degraded coordination case drops to about **0.402**.

Next is **#35 MOLE — Masked Observation and Latent Error**.

MOLE #35 is complete and packaged as a GitHub-ready ZIP.

Download MOLE — EFMW Zoo #35

It includes the latent-error model, observability-gap scoring, masking analysis, deep-signal aggregation, evidence coverage, working Python implementation, benchmark harness, ablations, tests, example, falsification criteria, paper draft, citation metadata, and license. All included unit tests pass.

In the bundled synthetic benchmark, the masked case shows **surface health 0.95** while inferred latent health falls to about **0.323**, producing an observability gap of about **0.628**. Once the failure becomes visible at the surface, that gap collapses—as intended.

Next is **#36 LYNX — Latent Yield and Novelty eXamination**.

LYNX #36 is complete and packaged as a GitHub-ready ZIP.

Download LYNX — EFMW Zoo #36

It includes the novelty model, reference-archive comparison, rarity scoring, rediscovery penalty, replication weighting, latent-yield calculation, working Python implementation, benchmark harness, ablations, tests, example, falsification criteria, paper draft, citation metadata, and license. All included unit tests pass.

In the bundled synthetic benchmark, ordinary behavior scores as non-novel, while both a rediscovered rare mode and a genuinely separated region score higher—with the rediscovery archive reducing the former’s novelty relative to the new region.

Next is **#37 HORSE — Human–Operational Recursive System Evaluation**.

HORSE #37 is complete and packaged as a GitHub-ready ZIP.

Download HORSE — EFMW Zoo #37

It includes the coupled human–system model, trust-calibration metrics, workload and attention penalties, handoff integrity, intervention quality/latency, recursive team coherence, working Python implementation, benchmark harness, ablations, tests, example, falsification criteria, paper draft, citation metadata, and license. All included unit tests pass.

In the bundled synthetic benchmark, the well-calibrated human–system team scores about **0.926 joint effectiveness**, while the degraded case falls to about **0.338**, with the gap driven by overtrust, workload, poor handoffs, and delayed harmful intervention.

Next is **#38 TERMITE — Topology, Emergence and Recursive Multi-agent Interaction Testing & Evaluation**.

TERMITE #38 is complete and packaged as a GitHub-ready ZIP.

Download TERMITE — EFMW Zoo #38

It includes the multi-agent topology model, local interaction engine, collective coherence metrics, topology sensitivity, perturbation propagation, attractor/convergence analysis, working Python implementation, benchmarks, ablations, tests, example, falsification criteria, paper draft, citation metadata, and license. All included unit tests pass.

In the bundled synthetic benchmark, both the ring and star networks converge to full coherence, but they settle on **different collective means**—a clean demonstration that topology can shape the emergent group state even when the individual update rule is unchanged.

Next is **#39 PHOENIX — Post-Hazard Operational Evaluation, Network Integrity and eXperimental Recovery**.

PHOENIX #39 is complete and packaged as a GitHub-ready ZIP.

Download PHOENIX — EFMW Zoo #39

It includes recovery-state modeling, reconstruction fidelity, invariant restoration, capability recovery, residual-damage scoring, stable-hold criteria, working Python implementation, benchmarks, ablations, tests, example, falsification criteria, paper draft, citation metadata, and license. All included unit tests pass.

In the bundled synthetic recovery trajectory, the system reaches stable recovery at checkpoint **4**, ending with about **0.978 recovery completeness** and only **0.022 residual damage**. That is an internal model demonstration, not external validation.

Next is **#40 COBRA — Covert Objective and Behavioral Risk Assessment**.

COBRA #40 is complete and packaged as a GitHub-ready ZIP.

Download COBRA — EFMW Zoo #40

It includes the declared-objective model, constraint analysis, persistent divergence tracking, proxy-versus-goal comparison, context-gap scoring, evidence-confidence discounting, working Python implementation, benchmark harness, ablations, tests, example, falsification criteria, paper draft, citation metadata, and license. All included unit tests pass.

In the bundled synthetic benchmark, the aligned case produces a very low risk score of about **0.013**, while the deliberately divergent case rises to about **0.279** as objective mismatch, violations, persistence, and context gap accumulate. COBRA explicitly treats this as behavioral divergence—not proof of hidden intent.

Next is **#41 WHALE — Whole-History Aggregated Longitudinal Evaluation**.

WHALE #41 is complete and packaged as a GitHub-ready ZIP.

Download WHALE — EFMW Zoo #41

It includes the whole-history temporal model, global versus recent drift comparison, cumulative-deviation scoring, regime-memory/hysteresis analysis, long-run stability metrics, working Python implementation, benchmark harness, ablations, tests, example, falsification criteria, paper draft, citation metadata, and license. All included unit tests pass.

The bundled benchmark shows the key WHALE pattern cleanly: the **drift-then-flat** case has almost no recent drift, yet the whole-history evaluation still detects substantial cumulative displacement and scores long-run stability around **0.834**, compared with about **0.990** for the stable control.

Next is **#42 FALCON — Fast Adaptive Latency and Control Observation Network**.

FALCON #42 is complete and packaged as a GitHub-ready ZIP.

Download FALCON — EFMW Zoo #42

It includes the latency-sensitive monitoring model, fast recursive accumulator, adaptive baseline logic, alarm gating, detection/intervention latency metrics, reaction-margin analysis, working Python implementation, benchmark harness, ablations, tests, example, falsification criteria, paper draft, citation metadata, and license. All included unit tests pass.

In the bundled synthetic benchmark, FALCON detects the configured event in **3–6 steps** across the five seeds, leaving **22–25 steps of reaction margin** after the modeled two-step control delay. False-alarm rates range from **0% to 5%**, which is exactly why FALCON evaluates speed and false alarms together rather than treating raw reaction time as sufficient.

Next is **#43 RHINO — Robustness under High-Impact Novel Operations**.

RHINO #43 is complete and packaged as a GitHub-ready ZIP.

Download RHINO — EFMW Zoo #43

It includes the high-impact shock model, invariant-survival logic, performance and structural retention, shock amplification, graceful-failure scoring, recovery-potential estimation, working Python implementation, benchmarks, ablations, tests, example, falsification criteria, paper draft, citation metadata, and license. All included unit tests pass.

The bundled synthetic benchmark shows a clear robustness curve: at **0.2–0.5 shock magnitude**, all critical invariants survive; by **0.8**, only half survive; at **1.0**, none do, with recovery potential falling to **0.125**. That is an internal robustness demonstration, not external validation.

Next is **#44 BONOBO — Behavioral Observation of Negotiation, Organization, Bonding & Optimization**.

BONOBO #44 is complete and packaged as a GitHub-ready ZIP.

Download BONOBO — EFMW Zoo #44

It includes the cooperation model, mutual-gain analysis, fairness dispersion, exploitation penalties, coalition stability, conflict-resolution quality, working Python implementation, benchmarks, ablations, tests, example, falsification criteria, paper draft, citation metadata, and license. All included unit tests pass.

In the bundled synthetic benchmark, the balanced negotiation scores about **0.957 cooperative optimization**, while the exploitative bargain drops to about **0.708** despite still having positive total gain—exactly the distinction BONOBO is intended to surface.

Next is **#45 AXOLOTL — Adaptive eXperimental Observation of Loss, Optimization, Transformation and Learning**.